INCH-POUND

MIL-H-24135/12(SH)
3 September 1993

MILITARY SPECIFICATION

HOSE, SYNTHETIC RUBBER, WIRE REINFORCED, HIGH PERFORMANCE, FOR FLEXIBLE HOSE ASSEMBLIES (SINGLE WIRE BRAID OVER SINGLE SYNTHETIC FIBER BRAID MEDIUM PRESSURE, HIGH TEMPERATURE, MULTI-FLUID, SIZES -4 THROUGH -40)

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This associated detail specification covers high performance synthetic rubber hoses, reinforced with a single layer of wire braid over a single layer of synthetic fiber braid in sizes -4 through -40 for medium pressure multi-fluid applications. Hose to this associated detail specification is compatible with both petroleum based and phosphate ester based oils. Hose to this associated specification sheet shall be colored blue.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 03Q42, 2531 Jefferson Davis Hwy, Arlington, VA 22242-5160 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4720

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SPECIFICATIONS

MILITARY

MIL-F-24787 - Fittings, End, Reusable for Flexible Hose Assemblies, General Specification for.
MIL-L-2104 - Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-H-5606 - Hydraulic Fluid, Petroleum Base; Aircraft Missile, and Ordnance.
MIL-L-17331 - Lubricating Oil, Steam Turbine and Gear, Moderate Service.
MIL-H-19457 - Hydraulic Fluid, Fire-Resistant, Non-Neurotoxic
MIL-H-24135 - Hose, Synthetic Rubber, Wire Reinforced, For Flexible Hose Assemblies, General Specifications for.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Hose furnished under this specification shall conform to the requirements of MIL-H-24135 and as specified herein.

3.2 Material.

3.2.1 Hose. Rubber compounds used shall be resistant to petroleum and phosphate ester based oils and sea water and shall consist of synthetic rubber or a blend of synthetic rubbers and shall meet the requirements of MIL-H-24135 and as specified herein.

3.3 Reinforcement. Reinforcement shall consist of a single layer of wire braid over a single layer of synthetic fiber braid applied in such a manner as to meet all the requirements of this associated detail specification.

3.4 Dimensions and pressures. Hose dimensions shall be in accordance with table I and hose pressures shall be in accordance with table II.

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TABLE I. Hose dimensions.

Hose size (dash no.)	Hose dimensions (inches)				Minimum bend radius (inches)
	ID tube		OD cover		
	min	max	min	max	
-4	0.188	0.214	0.500	0.539	3.00
-5	0.250	0.281	0.570	0.601	3.38
-6	0.313	0.344	0.656	0.695	4.00
-8	0.406	0.437	0.743	0.789	4.62
-10	0.500	0.539	0.910	0.945	5.50
-12	0.625	0.667	1.055	1.101	6.50
-16	0.913	0.940	1.250	1.290	7.38
-20	1.125	1.172	1.469	1.531	9.00
-24	1.375	1.422	1.719	1.781	10.50
-32	1.812	1.859	2.190	2.266	13.25
-40	2.375	2.437	2.828	2.922	24.00

TABLE II. Hose pressures and weight.

Hose size (dash no.)	Pressure (psi)		Impulse peak pressure (psi)	Approximate weight (lbs/ft)	Vacuum (inches of Hg)
	Working (max)	Burst (min)			
-4	3,000	12,000	3,750	0.15	28
-5	3,000	12,000	3,750	0.19	28
-6	2,250	9,000	2,813	0.23	28
-8	2,000	8,000	2,500	0.28	28
-10	1,750	7,000	2,188	0.39	28
-12	1,500	6,000	1,875	0.47	28
-16	800	3,200	1,000	0.41	20
-20	625	2,500	625	0.59	20
-24	500	2,000	500	0.72	15
-32	300	1,200	300	0.90	11
-40	300	1,200	300	1.63	11

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance provisions shall be in accordance with MIL-H-24135 and as specified herein.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Sampling for qualification testing.

4.2.1 Hose. Qualification testing of hose shall consist of hose samples as specified in table III. Samples shall be not less than 15 inches in length and shall be fitted with end fittings qualified to MIL-F-24787.

4.2.1.1 End fitting qualification. End fittings used for the qualification of hose to this associated detail specification shall be fittings qualified to group VIII, MIL-F-XX216. Where there are no previously qualified end fittings, the prospective hose supplier shall work together with a prospective fitting supplier to qualify their respective products. (The fitting supplier and hose supplier may be a single source.)

TABLE III. Hose samples required.

Hose size (dash no.)	No. of hose samples to be provided
-8	4
-32	5

4.2.1.2 The quantity of end fittings required shall be in accordance with table IV.

TABLE IV. End fittings required.

End fitting size (hose dash no.)	No. of end fitting samples to be provided
-8	8
-32	10

4.2.2 End fitting mix. Where there are two or more qualified suppliers of end fittings to group VIII of MIL-F-24787, the sample end fitting mix shall be in accordance with table V. Where there is only one previously qualified supplier, the "previously qualified" samples shall come from that supplier.

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TABLE V. Hose and fitting mix for combined qualification, or where there is more than one qualified fitting source.

One approved source					
Hose size (dash no.) 3/	No. of hose samples to be provided		End fitting 4/ (hose dash no.) 3/	No. of end fittings 4/ to be provided	
	Qualifier 2/	Appvd 1/ source		Qualifier 2/	Appvd 1/ source
-8	2	2	-8	4	4
-32	3	2	-32	5	5

Two or more approved sources			
Hose size (dash no.)	Number of hose samples to be provided by		
	Qualifier 2/	Approved 1/ source #1	Approved 1/ source #2
-8	2	1	1
-32	3	1	1
End fitting size (dash no.)	Number of end fittings to be provided by		
	Qualifier 2/	Approved 1/ source #1	Approved 1/ source #2
-8	4	2	2
-32	5	3	2

(Hose samples shall be not less than 15 inches in length and shall be coupled on one end with the fitting being submitted for qualification and on the other end with previously qualified end fittings.)

1/ Approved source - Previously qualified hose and fitting, or hose or fitting.
2/ Qualifier - Offeror of hose and fittings, or hose to be qualified.
3/ Satisfactory completion of qualification testing of the samples selected in accordance with tables III, IV and V will qualify the manufacturer for all sizes in this associated detail specification.
4/ End fittings shall be in accordance with MIL-F-24787.

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4.2.3 Hose and end fittings combined. In the event that a contractor for hose and end fittings or a hose contractor and a fitting contractor desire to qualify hose to this associated detail specification and fittings to group VIII, MIL-F-24787 in a combined qualification test, the samples required for testing shall be in accordance with table V.

4.2.4 Sample requirements when there are less than two previously qualified suppliers. Where there is only one qualified source for qualified samples, all the "previously qualified" samples required shall come from that source. Where there are no previously qualified source, all samples of hose and fittings shall be submitted by the prospective qualifier(s).

4.2.5 Sequence of tests on sample hose assemblies shall be as specified in table VI.

TABLE VI. Sequence of qualification tests on sample assemblies.

Test sequence 1 (See MIL-H-24135) 2 (See MIL-H-24135/12)	Sample sizes / numbers									
	Size -8					Size -40				
	1	2	3	4	5	6	7	8	9	10
Proof Pressure (see 4.6.1)$_1$	X	X	X	X	X	X	X	X	X	X
Stability test (see 4.6.2)$_2$	X	X	X	X	X	X	X	X	X	X
Aging test (see 4.6.3)$_1$					X				X	
Hot oil circulation (see 4.4.4)$_2$		X				X				
Impulse test (see 4.5)$_2$				X				X		
Burst test (see 4.6.6)$_1$	X									X
Vacuum test (see 4.6)$_2$					X					
Cold temperature flexibility (see 4.6.8)$_1$			X				X			

4.3 Test fluid. The test fluid shall be petroleum base oil in accordance with MIL-L-17331, MIL-L-2104 or MIL-H-5606 for the aging, hydraulic fluid circulation, impulse and cold temperature tests as specified herein.

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4.4 Hot oil circulation. Sample hose assemblies shall be filled with hydraulic fluid n accordance with MIL-L-2104, MIL-H-5606 or MIL-L-17331. Assemblies shall be installed in a 180 degree arc and tested on a flex and circulation test stand as shown on figure 1. The radius at the inside of the bend shall be equal to the applicable arc and flex bend radius specified in table I. Oil maintained at +300 ± 5°F shall be circulated through the assemblies at 60 psi while the assemblies are flexed 3.5 inches in each direction at 12 ± CPM. The hose assembly shall be subjected to 400 hours of hot oil circulation and flexing without evidence of leakage or other malfunction. The oil shall be changed after 200 hours of circulation.

4.5 Impulse pressure. The peak impulse pressure shall be 125 percent of the maximum rated working pressure specified in table II for sizes -4 through -16 and 100 percent of working pressure for sizes -20 through -40. The pressure time curve shall conform to figure 2. Impulse testing shall be conducted at 300°F and shall be for one million cycles. Size up to -16 shall be bent to 90 degrees and size -20 and larger shall be straight with one end free. The radius at the inside of the arc of the size up to -16 shall be equal to the minimum bend specified in table I. The rate of impulsing shall be 70 ± 35 CPM with the pressure curve conforming to figure 2. The test fluid shall conform to MIL-L-2104, MIL-H-5606 or MIL-L-17331. Heating or cooling of the test chamber is not permitted to maintain the required test temperature. For each size sample tested, failure to meet one of the following conditions shall constitute failure for that group:

(a) Failure of any one test assembly to attain 750,000 impulse cycles.
(b) Failure of any group to average one million impulse cycles.

(1) No average greater than 1,250,000 cycles shall be included.
(2) The cycles from no greater than six assemblies shall be included in the calculation.

4.6 Vacuum. Vacuum testing shall be performed both prior to and after oil aging. Sample size -4 through -16 shall be bent to 180 degrees and -20 through -40 to 90 degrees to the minimum bend radius specified in table I and the O.D. remeasured at the flattened section. Deformation of the O.D. in excess of 10 percent of the original O.D. shall be cause for rejection of the sample. With the hose samples held in the bent position, a vacuum as specified in table II shall be applied and held for 5 minutes. The hose shall be rechecked for additional flattening and for leakage as evidence by loss of vacuum. Application of the vacuum shall not result in greater than an additional 10 percent reduction in the flattened section of the O.D. for size up to and including -24, and 25 percent reduction for size -32 and larger. Upon completion of the vacuum test the sample assemblies shall be subjected to a 350 psi pressure test with no leakage from the hose or fittings permitted. The sample hoses shall then be dissected and examined. Any evidence of ply separation, blistering, collapse or other damage shall be cause for rejection.

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5. PACKAGING

5.1 Packaging requirements shall be in accordance with MIL-H-24135.

6. NOTES

(This section contains information of a general or ecplanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Hose to this specification is intended for use in shipboard or shore to ship piping systems where pressures do not exceed the rated working pressure, and temperatures do not exceed minus 55°F or plus 300°F.

6.2 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1 and 2.1.1).
(c) Size hose.
(d) Quantity of hose (linear feet) per size.
(e) Type and size of end fitting for hose assembly (see MIL-F-24787).

6.3 Subject term (key word) listing.

Group VIII
Phosphate esters
Petroleum
Sea water

Preparing activity:
Navy - SH
(Project 4720-N039-12)

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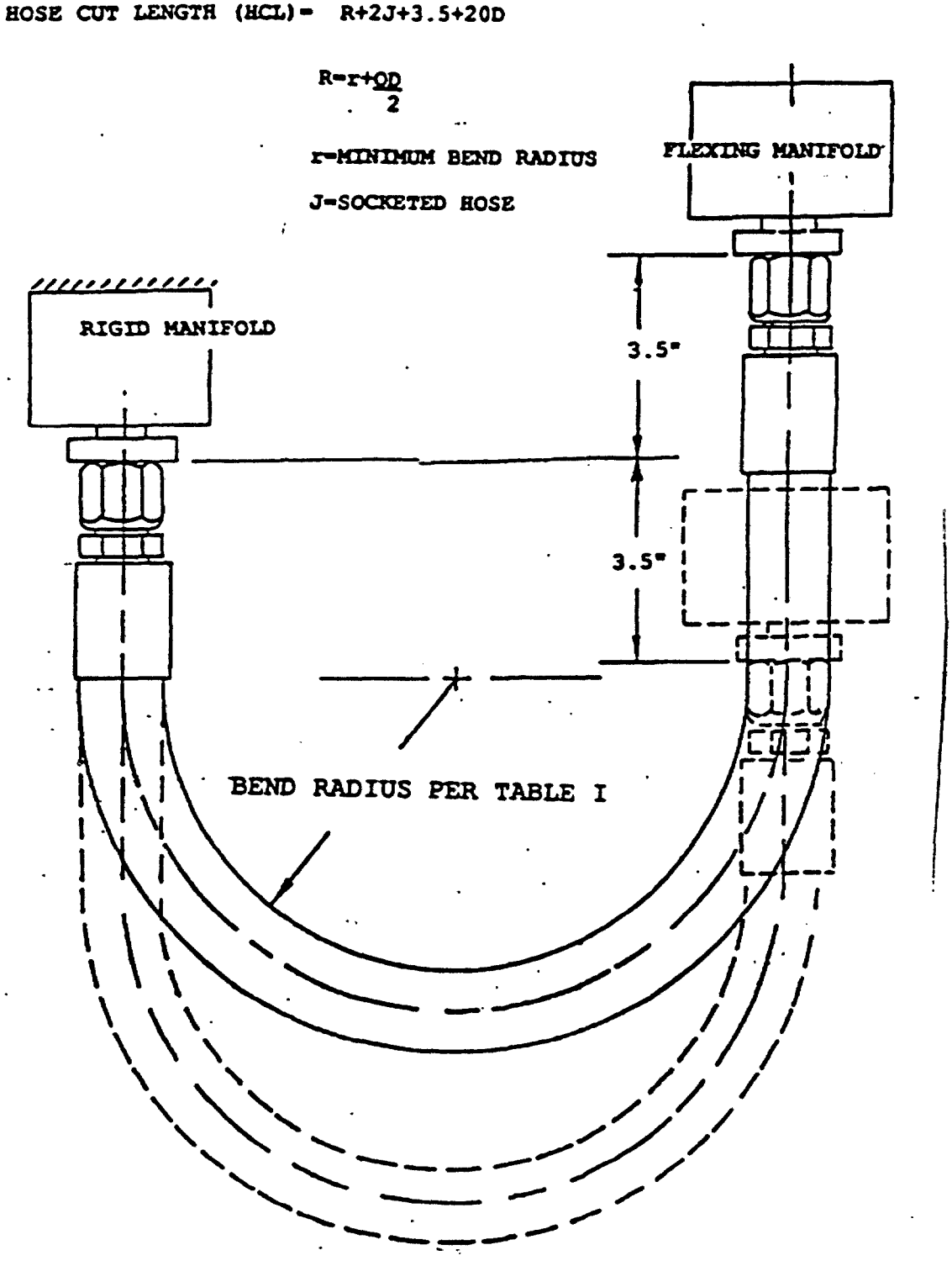


FIGURE 1. Flex and circulation test stand.

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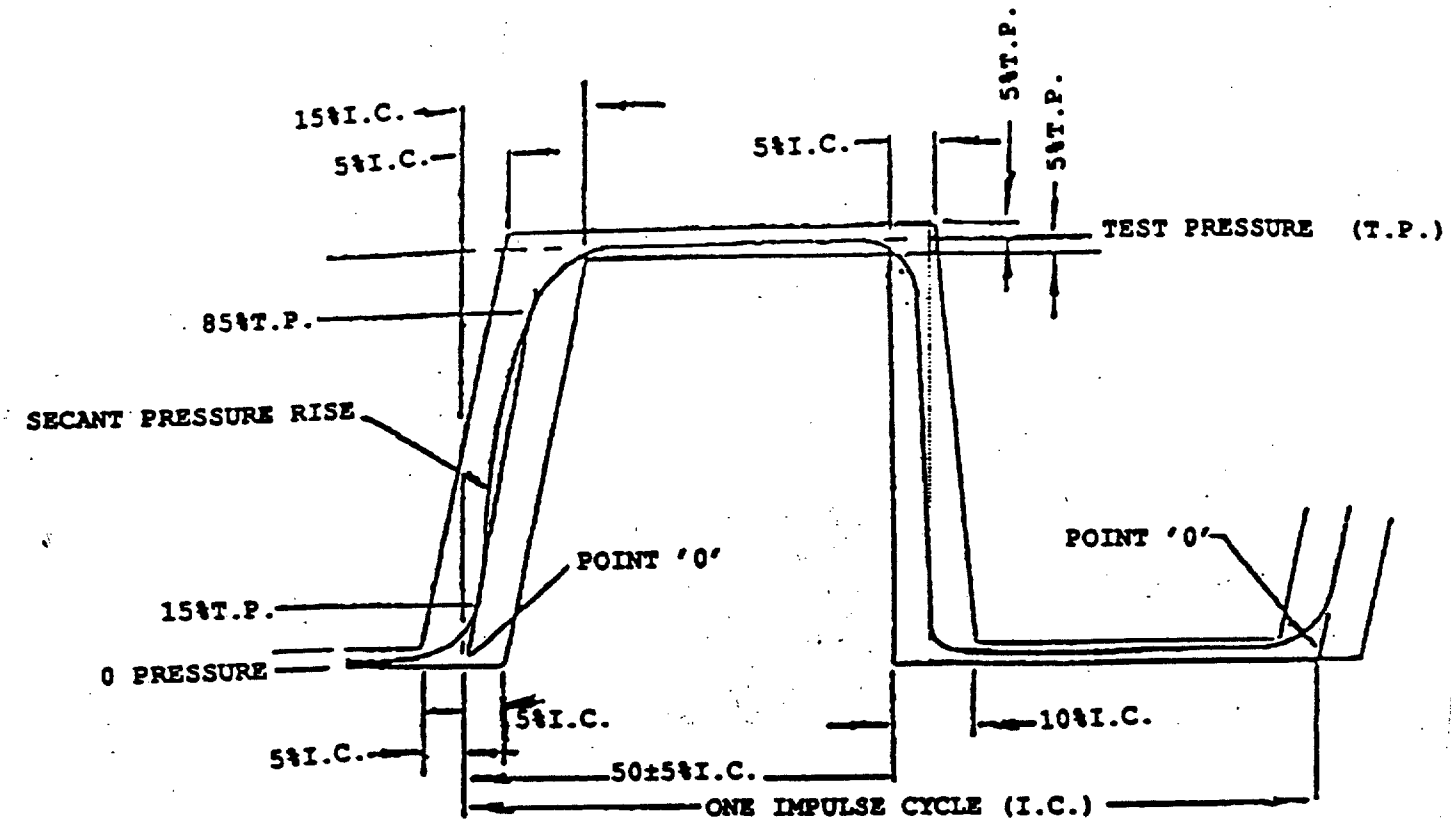


FIGURE 2. Impulse pressure curve.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the comment number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of this form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-H-24135/12(SH)	2. DOCUMENT DATE (YYMMDD) 930903

3. DOCUMENT TITLE
HOSE, SYNTHETIC RUBBER, WIRE REINFORCED FOR FLEXIBLE HOSE ASSEMBLIES (SINGLE WIRE BRAID OVER SINGLE SYNTHETIC FIBER BRAID MEDIUM PRESSURE, HIGH TEMPERATURE, MULTI-FLUID, SIZES -4 THROUGH -40)

4. NATURE OF CHANGE (identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial: (2) DSN (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Technical Point of Contact (TPOC) MR. CARL BANKS, SEA 03Y23 ADDRESS ALL CORRESPONDENCE AS FOLLOWS:	b. TELEPHONE (Include Area Code) (1) Commercial: DSN: TPOC: 703-602-0367 8-332-0367
c. ADDRESS (Include Zip Code) COMMANDER, NAVAL SEA SYSTEMS COMMAND ATTN: SEA 03Q42 2531 JEFFERSON DAVIS HIGHWAY ARLINGTON, VA 22242-5160	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403 Falls Church, VA 22041-3466 Telephone 703-756-2340 DSN 289-2340

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